

SC2207/CZ2007

Intro to Database Systems

Course Guide



SC2207/CZ2007

- Welcome to SC2207/CZ2007!
- This Course Guide describes the structure of the course; it serves as a roadmap for the different learning components of the course.
- This Course Guide contains important information regarding the course learning outcomes, learning materials, and assessment breakdown.

Course Instructors (full time students)

- Before recess week
- Dr. W. K. Ng (Course Coordinator)
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- After recess week
- Dr. Cong Gao
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Course Instructor (part time students)

- Entire semester
- Dr. Quah Tong Seng
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Course Objectives

- Database management systems (DBMS) are software systems that control the creation, maintenance, and use of databases (organized collections of data).
- DBMS are incredibly ubiquitous today; they underlie technology used by most people every day if not every hour.
- This course aims to equip you with the basics of modeling and constructing a relational database, querying and extracting data from the database.

Learning Outcomes

- Be able to model and design a relational database of medium complexity
- Be able to ensure data integrity through database normalization
- Be able to formulate database queries (relational algebra & SQL)
- Understand the usage of indexing to improve query efficiency
- Understand the significance of XML and JSON in today's world

Topics Overview

- Relational Database (7 weeks)
 - Modeling
 - Functional dependencies
 - Normalization
 - Querying using relational algebra
- SQL and Semi-structured data (5 weeks)
 - What it is?
 - Querying using SQL and Indexing
 - XML-to-Relational mapping
 - JSON, NOSQL

Topics for Week 1 to Week 7

- **Topic 1:** Entity-relationship modeling (ER)
- **Topic 2:** Functional dependencies (FD)
- **Topic 3:** Boyce-Codd Normal Form (BCNF)
- **Topic 4:** Third Normal Form (3NF)
- **Topic 5:** Relational algebra

JANUARY						
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				4		
	5	6	7	8	9	10
1	12	13	14	15	16	17
2	19	20	21	22	23	24
3	26	27	28	29	30	31

Teaching Week

Topic 1

Topic 1 and 2

Topic 2

FEBRUARY						
	S	M	T	W	T	F
						1
4	2	3	4	5	6	7
5	9	10	11	12	13	14
6	16	17	18	19	20	21
7	23	24	25	26	27	28

Teaching Week

Topic 3

Topic 4

Topic 5

Topic 5

Topics for Week 8 to Week 12

- SQL and Indexing
- XML and JSON
- More details on weekly schedule to be provided later

Course Components

- Lectures (12 weeks, 2 hours per week)
- Tutorials (10 weeks, 1 hour per tutorial)
- Lab project (5 sessions, 2 hours per session)

Lectures (12 weeks)

- Start from Week 1
- Two lectures per week
- Students are encouraged to go through slides before coming to lectures.

Tutorials (10 weeks)

- Start from Week 3
- Each tutorial consists of two sets of questions
 - Practice exercises
 - Answers will be released in the tutorials
 - Critical thinking exercises
 - Optional
 - Answers will not be released
 - Come up with your own solutions, then liaise with your tutor for further discussions

Lab Project (5 sessions)

- Start from Week 3
- Hands-on practice on design and implementation of a real-world relational database application
- Lab manual available on NTULearn SC2207/CZ2007 website
- Only reports from Labs 1, 3 and 5 will be graded.
- Attendance will be taken for Labs 1, 3, and 5.

Quiz

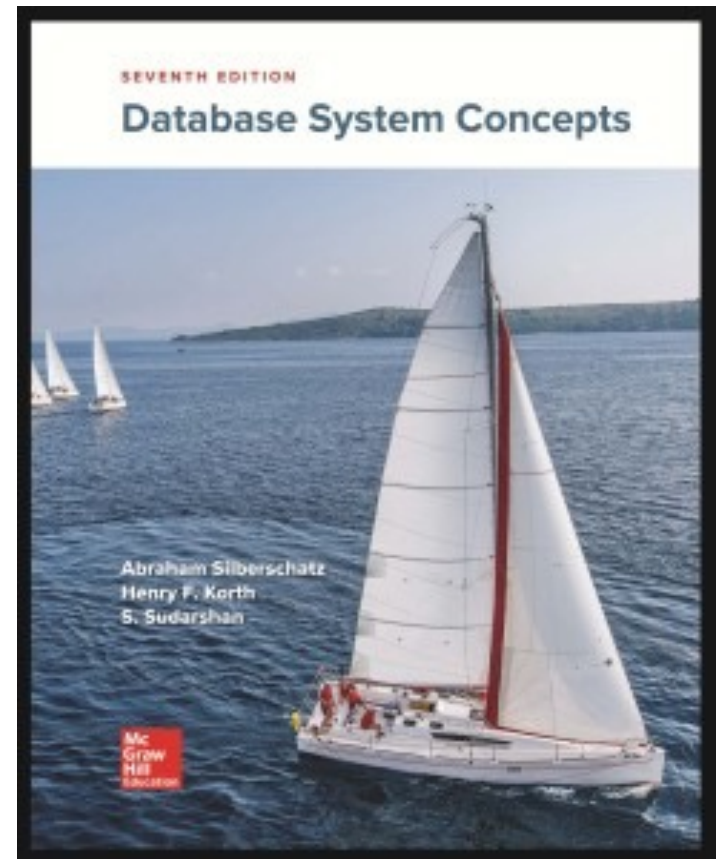
- To hold on 27 Feb 2025 during scheduled lecture session
- Quiz covers Topics 1, 2, 3, 4
- Quiz mode: closed book, MCQ, online, 45 mins, bring your own labtop/tablet (ensure sufficient battery life)
- **Don't miss the quiz**
 - Missing quiz = zero mark, if not on MC or LOA

Course Assessment

- Lab project: 40%
 - (Lab1: 10%, Lab3: 10%, Lab5: 20%)
- Quiz: 10%
- Exam: 50%
- Percentage subject to change

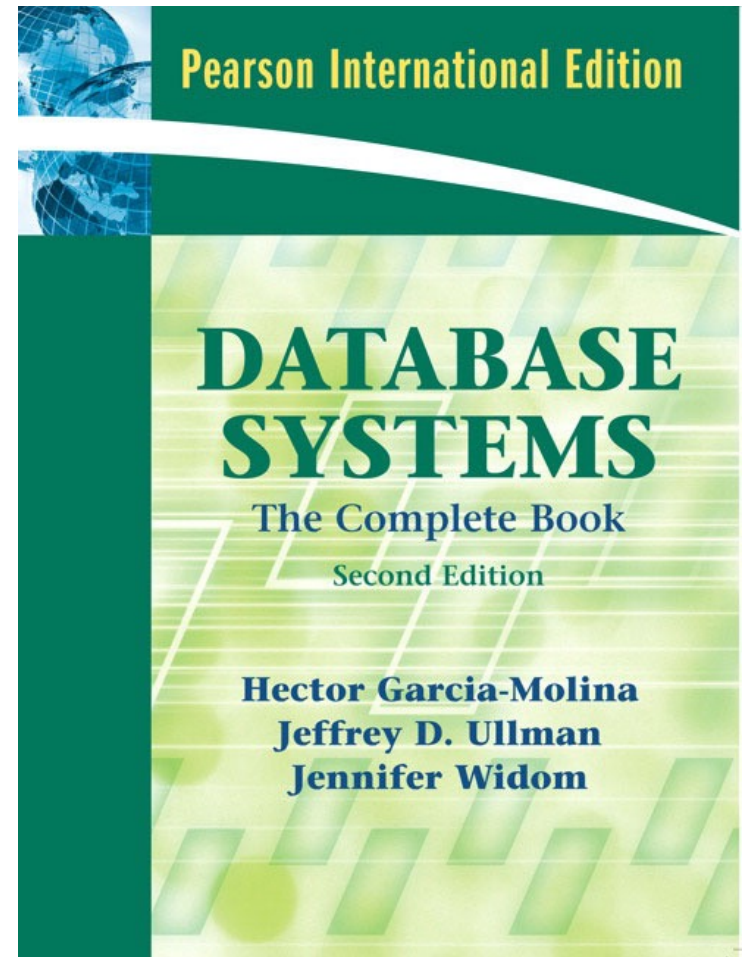
Learning Materials

- Database System Concepts
7th Edition (2019)
- Avi Silberschatz, Henry F. Korth, S. Sudarshan
- McGraw-Hill
- ISBN 9780078022159



Learning Materials

- Lecture slides are sufficient for learning
- For additional (optional) reading, students may refer to “Database Systems: The Complete Book”



A hand is holding a white rectangular sign against a bright blue sky filled with white, fluffy clouds. The sign is tilted slightly to the right. The text on the sign is written in a black, casual, handwritten style. The word 'Your' is on the first line, 'Journey' is on the second line, and 'Starts Here' is on the third line. The hand is visible on the left side of the sign, with fingers gripping the edge. The overall image has a positive, motivational feel.

Your Journey
Starts Here